

Abbas Rizvi

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EDUCATION

Bachelor of Science; Computer Science

Expected Graduation: 2028

University of Alberta, Edmonton, AB

GPA: 3.67

SKILLS

Programming C, C++, Python, Java, JavaScript

Data & ML: PyTorch, XGBoost, NumPy, Pandas, Matplotlib, SPSS, Excel, YOLO, OpenCV

Web & Tools: Flask, Node.js, HTML, Git, HPC, AWS (S3, RDS, ECS, ECR, SageMaker), Docker

EXPERIENCE

Machine Learning Developer | *Technologies:* Python, PyTorch, YOLO, OpenCV, AWS, Docker, Numpy, PostgreSQL

ChromacareLabs | Dec 2025 - Present

- Built a comprehensive ML pipeline from CV-based feature extraction through deep neural network architecture design, training, and evaluation for regression tasks, for production deployment on AWS.
- Managed large-scale datasets (HDF5, PostgreSQL on RDS, S3) via relational schema for training data and metadata; built API endpoints and visualization tooling for data collection and model testing.

ML Research Assistant, Servier Virtual Cardiac Centre | *Technologies:* Python, PyTorch, NumPy, Matplotlib, Pandas, Git, HPC

University of Alberta | May 2025 – Present

- **First author** on a paper (IEEE BIBE 2025) after benchmarking deep learning models for cardiac MRI segmentation, demonstrating improved accuracy with transfer learning.
- Applying regression models and numerical integration to clinician-reviewed data, which estimates ventricular volumes from fewer short-axis MRI slices, suggests the potential to significantly shorten scan times.

Teaching Assistant, CMPUT 201 – Practical Programming Methodology

University of Alberta | Expected Sept 2025 – Dec 2025

- Guide students in labs, office hours, and forums on C programming and developing well-structured programs, support Git version control and terminal usage, and assist with assignment creation.
- Evaluate assignments and exams, provide detailed feedback to help students improve programming and problem-solving skills, and ensure grading is consistent and aligned with course standards.

Data Analysis Assistant

Faculty of Medicine & Dentistry, University of Alberta | Feb 2025 – Aug 2025

- **Co-authored** a research article and a poster on pain assessment in dementia, presented at the Canadian Conference on Dementia, a leading national conference, by contributing data analysis and visualizations using SPSS and Excel.

SOFTWARE PROJECTS

3D DICOM Cardiac MRI Visualizer | *Technologies:* C++, VTK, Qt

- Developed a C++/VTK/Qt tool to construct DICOM volumes from 2D slices using slice position, spacing, and orientation metadata, while mapping contours of anatomical structures onto the reconstructed volume.
- Enabled interactive 3D visualization of DICOM time series, letting users select slices and view volumetric changes.
- Will use generated 3D visualizations to explain methodology in a research paper.

Barnes-Hut N-Body Simulation in C | *Technologies:* C, Python, Git, Matplotlib, multiprocessing

- Engineered a high-performance C physics simulation for gravitational N-body systems, reducing pairwise interaction complexity from $O(n^2)$ $\rightarrow$ $O(n \log n)$ via the Barnes-Hut algorithm.
- Built a recursive quadtree to partition 2D space, dynamically subdividing dense regions for faster force calculations.
- Developed a Python visualization pipeline, parallelizing thousands of frames and simulating collisions of three galaxies.

City of Edmonton Property Value Predictor | *Technologies:* Python, XGBoost, Sklearn, Pandas, NumPy, Flask, Render

- Collected and integrated property data for 360,000 Edmonton residences from open data sources and OpenStreetMap, processing physical and location-based features.
- Trained a 5-fold XGBoost model ($R^2 = 0.917$, MAE = \$21.5K, RMSE = \$83.4K) and deployed as a Flask app on Render.